Forecasting German Electricity Demand

A Comparative Study of Statistical, Machine-Learning and Deep-Learning Approaches

Ram Sai Sayani – 24080366 / *MSc Data Science — Advanced Research Topics, Assignment 1*

GitHub Link: https://github.com/Ram-sai7/electricity-demand-forecasting.git

1. Introduction

Electricity system operators and utilities need reliable demand forecasts at horizons ranging from a few hours to several years, since generation scheduling, reserve procurement and long-term capacity planning all depend on knowing how much load the grid will be asked to serve. This report develops and compares five families of forecasting model — simple statistical benchmarks, a seasonal ARIMA model, a temperature-conditioned SARIMAX model, feature-based tree ensembles, and a recurrent neural network — applied to German national electricity load from the Open Power System Data (OPSD) time series package (Open Power System Data, 2020), covering hourly load from 1 January 2015 to the end of the available file in October 2020. Weekly mean temperature for Berlin, taken as a representative German location, was retrieved from the Open-Meteo historical archive (Open-Meteo, 2024) and used as an explanatory covariate in Part 4 of the analysis.

The guiding question throughout is not simply "which model has the lowest error", but why each model performs the way it does given the choices made in building it — the differencing orders chosen for SARIMA, the way lag features were engineered for the tree ensembles, and the sequence length and rollout strategy used for the LSTM. Section 2 covers the exploratory analysis and formal stationarity testing that motivated the SARIMA specification. Section 3 briefly describes each modelling approach and the reasoning behind using it. Section 4 presents the results, evaluated on a held-out two-year (104-week) forecast horizon running from October 2018 to October 2020. Section 5 offers a critical discussion of why the models performed as they did, including two results that ran contrary to my initial expectations. Section 6 answers the set questions directly, Section 7 discusses limitations and future work, and Section 8 concludes.

2. Data and Exploratory Analysis

The raw OPSD file provides hourly German load (DE_load_actual_entsoe_transparency) in megawatts. After restricting to 1 January 2015 onward, the cleaned hourly series contains 50,400 observations running from 2015-01-01 to 2020-09-30. This was aggregated to a daily series (2,100 observations) and a weekly series (301 observations, converted to gigawatts for readability) using simple mean resampling, with any small resulting gaps closed by time-based linear interpolation.

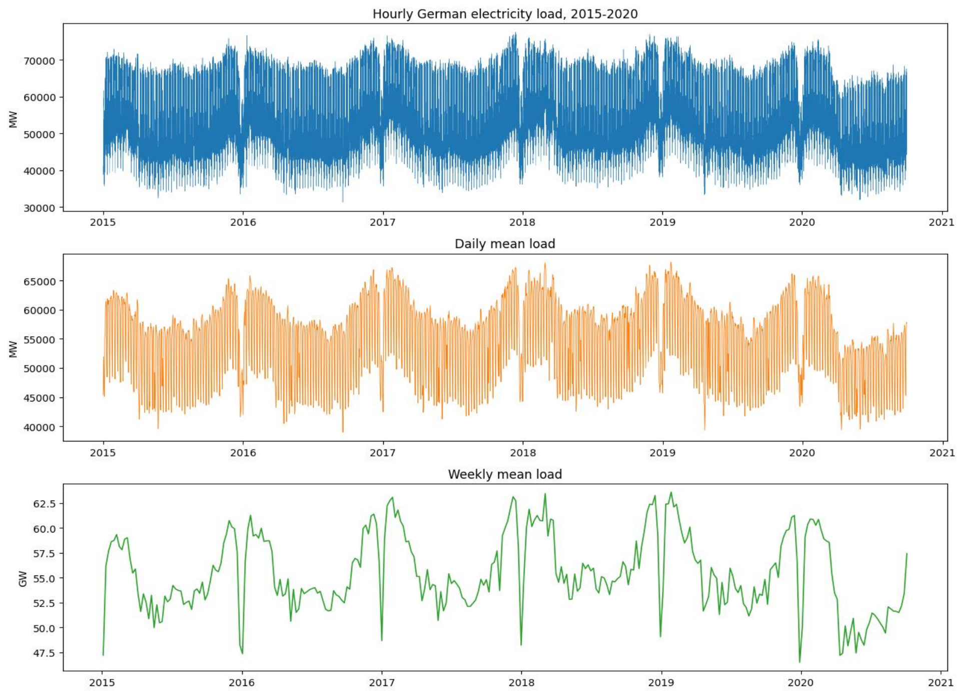

Figure 1. German electricity load at hourly, daily and weekly resolution, 2015–2020.

Figure 1 shows the expected structure at each resolution: the hourly series is dominated by intraday cycling that is invisible once the data is aggregated to daily or weekly means, while the weekly series makes the underlying annual seasonal pattern — higher load in winter, a secondary summer dip around the holiday period — much easier to read. A visible, if irregular, dip appears toward the end of the series in 2020, consistent with the demand shock associated with COVID-19 lockdown measures; this is a genuine structural break rather than a data artefact, and its consequences for the forecast evaluation are discussed in Section 5.

2.1 Seasonal decomposition

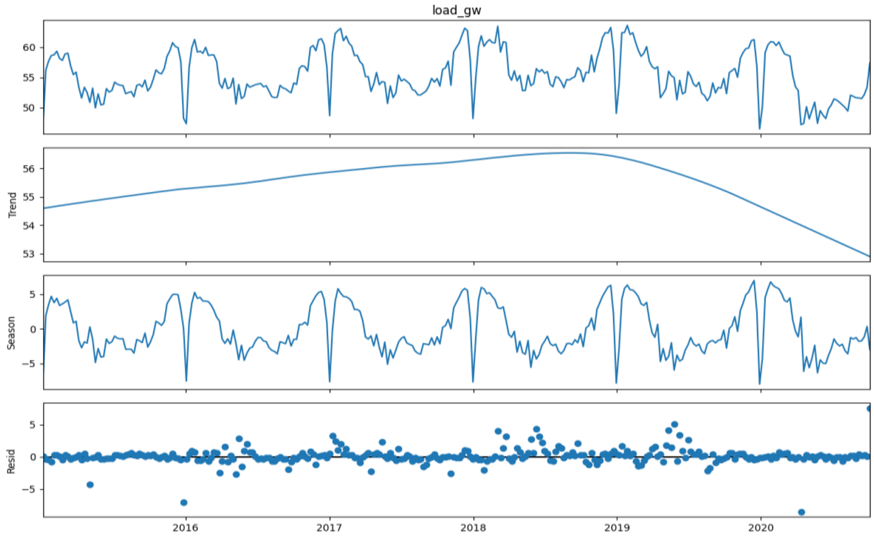

Figure 2. STL decomposition of the weekly load series (annual period, s = 52).

An STL decomposition (Cleveland et al., 1990) of the weekly series with an annual period gives a seasonal strength of 0.871 against a trend strength of only 0.383. In plain terms: the annual seasonal cycle explains the overwhelming majority of the systematic variation in weekly load, while the underlying trend component is comparatively weak and, as Figure 2 shows, not monotonic — it drifts gently rather than growing or shrinking in a fixed direction. This single number was the main justification for treating the problem as a seasonal one throughout — fitting SARIMA rather than plain ARIMA, and giving the tree ensembles an explicit lag-52 feature rather than relying on them to discover the annual cycle unaided.

2.2 Stationarity testing

Two complementary tests were used, since they test opposite null hypotheses: the Augmented Dickey-Fuller (ADF) test, whose null is that the series has a unit root, and the KPSS test, whose null is that the series is stationary. Both were applied to the level series, the first difference, the seasonal (lag-52) difference, and the combination of both, following the approach in Box et al. (2015) and Hyndman and Athanasopoulos (2021).

Series	ADF stat	ADF p	KPSS stat	KPSS p	Verdict
Level	-4.047	0.0012	0.160	0.10	Ambiguous (see discussion)
First difference (d=1)	-7.069	<0.0001	0.056	0.10	Stationary
Seasonal difference (D=1, s=52)	-4.295	0.0005	-	-	Stationary
First + seasonal difference	-7.278	<0.0001	-	-	Stationary

Table 1. Stationarity test results for the weekly load series.

The differenced series behave exactly as textbook theory predicts: both the ordinary first difference and the combined first-and-seasonal difference pass ADF comfortably ($p < 0.001$) and are not rejected by KPSS. The level-series result, however, is genuinely ambiguous — ADF actually rejects the unit-root null at the level ($p = 0.0012$), which taken at face value would suggest the raw series is already stationary, contradicting the clear seasonal swings visible in Figure 1. My interpretation, discussed further in Section 5, is that ADF has limited power against a strongly seasonal alternative: a series that cycles regularly around a roughly constant annual mean can look locally mean-reverting to the test even though it is not stationary in the sense that matters for forecasting. I therefore did not rely on the level-series ADF result in isolation: the autocorrelation function of the level series retains a persistent, significant spike at lag 52 that a first difference alone does not remove, which — together with the STL seasonal strength of 0.871 — was the primary justification for the seasonal difference used in the SARIMA specification.

3. Methodology

Four families of model were fitted and evaluated on the same held-out two-year (104-week) test period from 14 October 2018 to 4 October 2020 using a chronological train/test split. Feature-based models used a rolling one-week-ahead evaluation, whereas the statistical and LSTM models were assessed over the full forecast horizon.

3.1 Benchmark forecasts

Mean, naive, seasonal-naive and drift forecasts (Hyndman and Athanasopoulos, 2021) were included not because they are competitive models in their own right, but because they set the bar that any more complex model needs to clear to justify its added cost. Seasonal naive in particular — simply repeating the value from the same week one year earlier — is a strong benchmark on a series with seasonal strength as high as 0.871, and, as Section 4 shows, this expectation was borne out.

3.2 SARIMA and SARIMAX

SARIMA was chosen over plain ARIMA specifically because of the seasonal-strength result in Section 2.1: an annual seasonal component this dominant cannot be captured by a non-seasonal model without an enormous number of ordinary AR/MA terms. Orders were selected by grid search over $p \in [0,6]$, $d \in [0,2]$, $q \in [0,6]$, with the seasonal difference fixed at $D = 1$ (justified in Section 2.2) and a restricted seasonal search over $(P, Q) \in \{0,1\}^2$, minimising AIC. A full 6-dimensional seasonal-and-non-seasonal grid was not computationally feasible — even the restricted search of 588 candidate models took just under five hours to fit, since every candidate involves a Kalman-filter state space with a 52-period seasonal component. SARIMAX extends the selected SARIMA order with weekly temperature, heating-degree-days and cooling-degree-days as exogenous regressors, to test whether a known external driver of demand improves on the purely autoregressive model.

3.3 Feature-based regression

A supervised-learning table was constructed from lagged load (1, 2, 4, 8, 13, 26 and 52 weeks), rolling means and standard deviations of load, lagged temperature, and Fourier/calendar terms for week-of-year, all built using shift operations so that no feature for week t uses information from week t or later. Random Forest (Breiman, 2001) and Histogram Gradient Boosting (Friedman, 2001) were chosen because tree ensembles handle non-linear interactions — for instance, temperature affecting load differently in winter than in summer — without requiring them to be specified explicitly, unlike SARIMAX, and because feature importances give at least an approximate account of what is driving the forecast.

3.4 LSTM

An LSTM (Hochreiter and Schmidhuber, 1997) was applied to the hourly series, motivated by its established use in short-term load forecasting (Kong et al., 2019), where it can in principle learn daily and weekly structure directly from a raw sequence without hand-engineered lag features. A one-week (168-hour) sliding window was used to predict the next hour; three architectures (varying hidden units, depth and dropout) were compared on a validation split carved from the training period, and the best configuration (32 units, one layer, no dropout) was retrained with early stopping. To reach the full two-year test horizon, the model was rolled forward iteratively — each hourly prediction was fed back in as part of the input for the next step — since the model was trained to predict only one step ahead.

4. Results

4.1 Benchmark forecasts

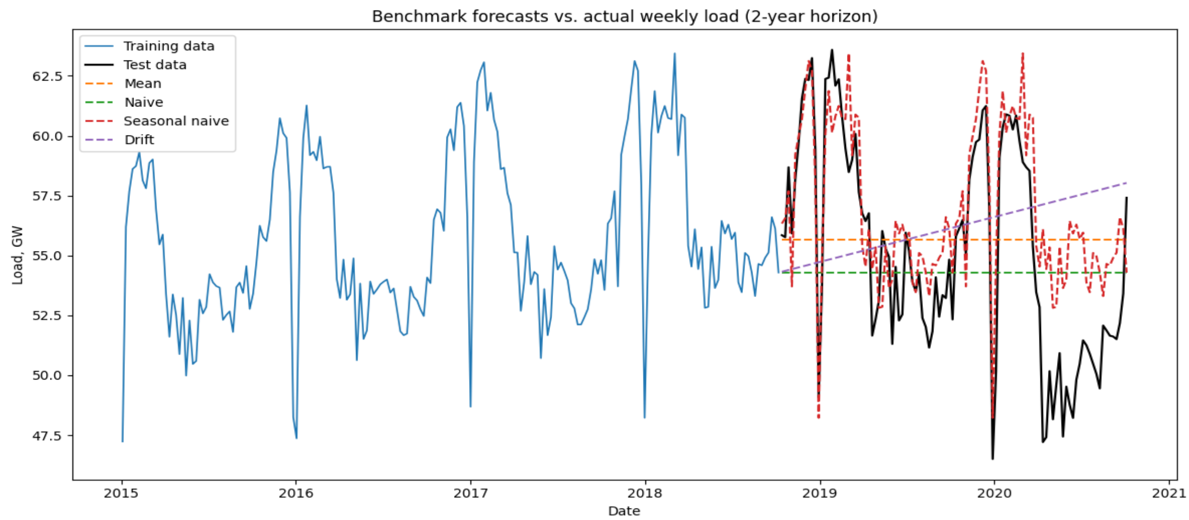

Figure 4. Benchmark forecasts against the two-year held-out test period.

Seasonal naive (MASE 1.732) clearly outperformed mean (2.831), naive (2.827) and drift (3.243), confirming the expectation from the seasonal-strength analysis: because most of the systematic variation in the series is the annual cycle itself, a forecast that simply repeats last year's value captures most of what there is to capture, and the other three benchmarks, which ignore seasonality altogether, are left a long way behind.

4.2 SARIMA

The AIC-minimising order from the restricted grid search was SARIMA(2,1,6)×(0,1,1,52), with AIC = 300.94. The fitted model, however, came with a convergence warning, and all six of its MA coefficients had standard errors several times larger than the coefficient estimates themselves (e.g. ma.L1 = 0.930, SE = 6.611, p = 0.888) — a strong sign that the model is over-parameterised for a training sample of only 197 weekly observations, and that AIC was rewarding a small in-sample fit improvement that did not correspond to genuinely useful structure.

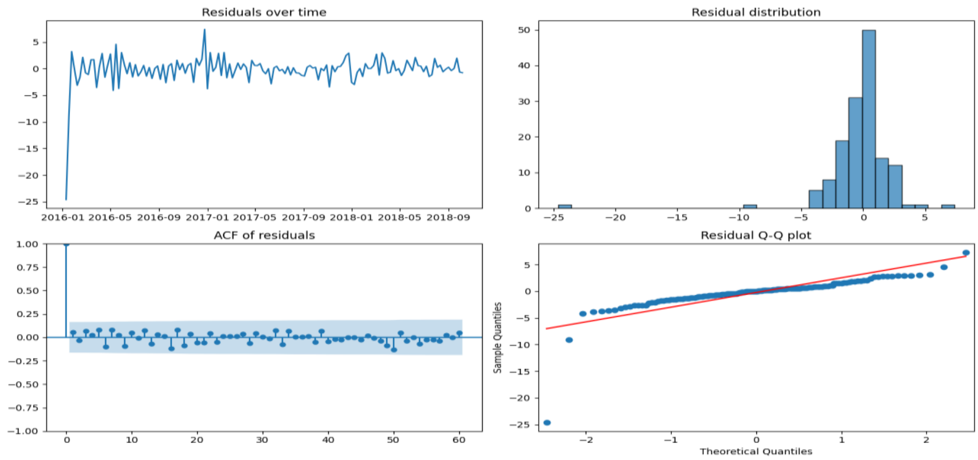

Figure 5. SARIMA residual diagnostics: residuals over time, distribution, ACF, and Q-Q plot.

Despite the estimation concerns, the residual diagnostics in Figure 5 are genuinely clean: the Ljung-Box test found no significant autocorrelation at lags 10, 20 or 52 (p = 0.757, 0.846, 0.997), and the Jarque-Bera test did not reject normality (p = 0.68). In isolation, this diagnostic picture would normally be read as evidence of a well-specified model. The gap between "residuals look like white noise on the training data" and "the model forecasts well out of sample" turned out to be the central lesson of this section.

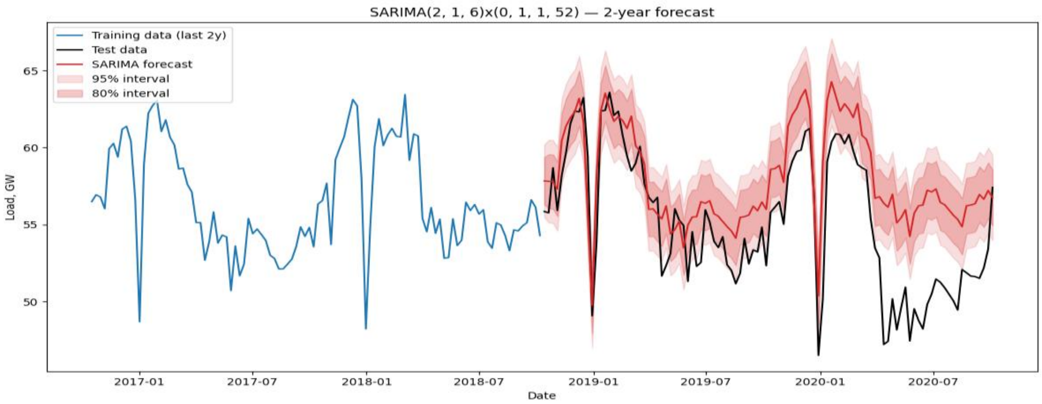

Figure 6. SARIMA two-year forecast with 80% and 95% confidence intervals, against actual load.

On the test period, SARIMA achieved MASE 2.301 — worse than seasonal naive, and with a mean positive bias of 2.800 GW, meaning it systematically over-predicted load throughout the forecast horizon (visible in Figure 6, where the forecast sits above the actual series for most of the window). This is consistent with the over-parameterisation concern above: a flexible model fit tightly to 2015–2018 extrapolated patterns from that window — including, most likely, a growth trend that did not continue — into a test period that both grew more slowly than the training years and, in its final months, included the COVID-19 demand shock that no model trained on pre-2020 data could have anticipated.

4.3 SARIMAX with temperature

Weekly load correlates with weekly mean Berlin temperature at r = -0.636 and with heating degree-days at r = +0.687, both in the expected direction for a heating-dominated demand profile. Adding temperature, heating- and cooling-degree-days as exogenous regressors to the same SARIMA order improved MASE from 2.301 to 2.146 and reduced the mean bias from 2.800 to 2.591 GW.

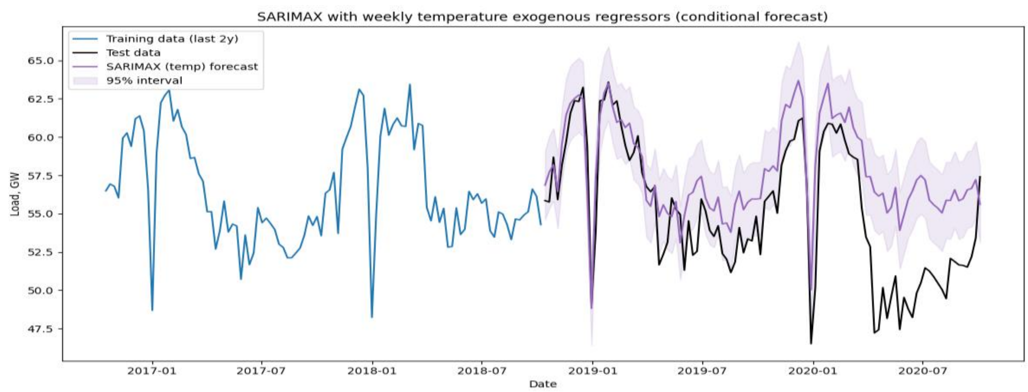

Figure 7. SARIMAX (temperature-conditioned) two-year forecast.

The improvement is real but modest, and SARIMAX still did not beat seasonal naive. Because the same over-parameterised (2,1,6)×(0,1,1,52) order underlies this model, the extra explanatory information from temperature was working on top of an already biased base forecast rather than fixing the source of that bias. It is also worth restating explicitly that this is a conditional, not an operational, forecast: it uses the temperature that was actually observed during the test period, not a forecast of it, so the reported accuracy overstates what would have been achievable in real time.

4.4 Feature-based regression

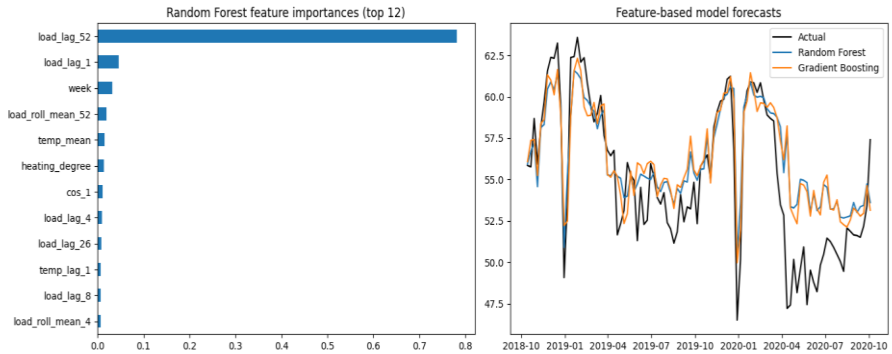

Figure 8. Random Forest feature importances (left) and forecasts against actual load (right).

Random Forest (MASE 1.405) and Gradient Boosting (MASE 1.411) were the only two models to beat seasonal naive, by a clear margin. The feature-importance ranking in Figure 8 shows load_lag_52 as the dominant predictor by a wide margin, followed by the shorter lags and the rolling-mean features, with temperature contributing a smaller but non-trivial share. This ranking is a reasonable explanation for why these models succeeded where SARIMA did not: they get the benefit of an explicit seasonal anchor — functionally similar to what seasonal naive already exploits — while additionally being able to make small, non-linear corrections around that anchor using recent load levels and weather, without being forced through a single global linear/ARMA parameterisation that has to fit the whole series at once.

4.5 LSTM

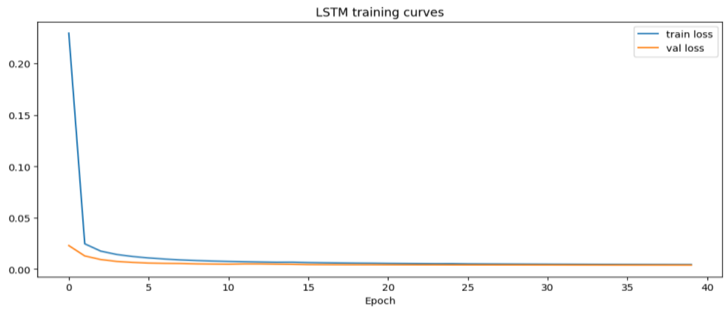

Figure 9. LSTM training and validation loss curves.

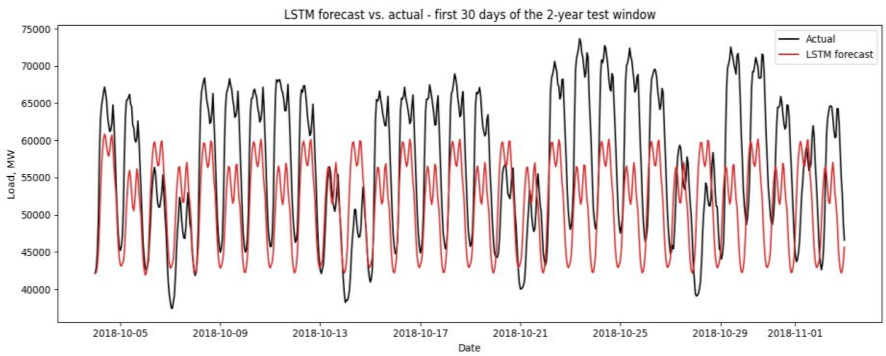

Figure 10. LSTM iterative forecast against actual hourly load, first 30 days of the test window.

Training itself was well behaved — Figure 9 shows validation loss tracking training loss with no obvious overfitting within the training period. The forecast, however, was the worst of any model tested once aggregated to weekly resolution: MASE 3.319, roughly 1.9 times seasonal naive's error, and marginally behind even the Drift benchmark (MASE 3.243). Figure 10 shows why: over the first 30 days of the two-year test window, the iteratively rolled-forward forecast tracks the broad level of actual load reasonably well at first but progressively loses the amplitude of the daily cycle, settling into a damped oscillation that under-predicts the peaks and drifts away from the true intraday pattern. Because each hourly prediction is fed back in as an input for the next step, small errors compound across the full 17,472-hour horizon, and by the later parts of the window the model is effectively forecasting from its own smoothed, accumulated drift rather than from real information about the system.

4.6 Summary of results

Model	MAE (GW)	RMSE (GW)	MASE	Bias (GW)
Random Forest	**1.938**	**2.611**	**1.405**	**1.134**
Gradient Boosting	**1.945**	**2.621**	**1.411**	**1.139**
Seasonal Naive	2.319	3.007	1.732	1.732
SARIMAX (temperature, conditional)	2.873	3.601	2.146	2.591
SARIMA	3.080	3.763	2.301	2.800
Naive	3.783	4.459	2.827	-0.882
Mean	3.789	4.397	2.831	0.481
Drift	4.340	5.118	3.243	1.007
LSTM (hourly, aggregated to weekly)	4.441	5.589	3.319	-3.484

Table 2. Evaluation metrics for all models, ranked by MASE (lower is better). MASE is scaled by the in-sample seasonal-naive error, so values below 1.0 would indicate an improvement over the training-period seasonal-naive benchmark; values below seasonal naive's own test-period MASE (1.732) indicate a genuine out-of-sample improvement over it.

5. Critical Discussion

5.1 Why the results turned out the way they did

Two results ran against my expectations and are worth summarising here, with full detail deferred to Section 6 (Q1 and Q5) to avoid repetition. First, SARIMA and SARIMAX underperformed seasonal naive because the AIC-selected order (six MA terms, a convergence warning, several insignificant coefficients) was over-parameterised for a 197-observation training sample — AIC rewarded a marginal in-sample fit improvement that did not generalise. Second, the LSTM's weak result stems from the iterative one-step rollout used to reach the two-year horizon: prediction errors compound across roughly 17,500 sequential hourly steps, which is a methodological/architectural issue rather than evidence that LSTMs are unsuitable for load forecasting generally. By contrast, Random Forest and Gradient Boosting succeeded largely because their dominant feature, load_lag_52, functions much like seasonal naive itself, while additional lags, rolling statistics and temperature allow small local corrections on top of that seasonal anchor.

5.2 Forecasts against the data actually observed

The two-year test window (October 2018–October 2020) is itself real data collected after the modelling period relative to the training window it was held out from, and it includes an out-of-sample event no training data could have anticipated: the COVID-19 demand shock visible as an irregular dip toward the end of Figure 1. Every model's positive bias in Table 2, bar the plain naive forecast, is consistent with this — models trained on 2015–2018 data extrapolated a level of demand that 2020 then undercut. This is less a modelling failure than a reminder that no model trained purely on historical patterns can anticipate a structural break; a fair operational evaluation would either exclude 2020 from the test window or treat the resulting error inflation as expected rather than diagnostic.

5.3 Interpretability, complexity and covariate availability

A fuller comparison of interpretability and complexity across SARIMAX, the feature-based models and the LSTM, and a full discussion of covariate availability, is given as a direct answer to Q5 and Q4 in Section 6, so is not repeated here in full. In brief: SARIMAX's apparent interpretability advantage was undermined in this study by a poorly identified fit; the tree ensembles offered a reasonable middle ground via feature importances; and the LSTM was both the least interpretable and least accurate model tested — a useful caution against defaulting to the most complex available architecture.

6. Answers to the Set Questions

This section answers the six set questions directly. Fuller supporting evidence for each point is developed earlier in the report, referenced below by section.

Q1. Compare all models against the seasonal-naive benchmark. Which, if any, provide a meaningful improvement?

Table 2 (Section 4.6) shows only two models beating seasonal naive (test MASE 1.732): Random Forest (MASE 1.405, a 19% reduction) and Gradient Boosting (MASE 1.411, an 18% reduction). Every other model — SARIMA, SARIMAX, the remaining benchmarks, and the LSTM — was worse than seasonal naive on this test window, some considerably so. This was not the outcome I expected before running the analysis; I had anticipated SARIMAX in particular to improve on seasonal naive given its access to both autoregressive structure and a weather covariate. Section 5.1 traces this to the AIC-selected SARIMA order being over-parameterised (six largely insignificant MA terms, a convergence warning), which let it fit the training data closely without generalising well, whereas the tree ensembles succeeded by leaning on a lag-52 feature functionally similar to seasonal naive itself, combined with smaller corrective signals from recent load and temperature.

Q2. How was data leakage avoided when creating temperature lag features?

Three safeguards were applied throughout the analysis to minimise data leakage. First, all lagged and rolling-window features were constructed using. shift(k) with k >= 1, ensuring that each feature for week t depended only on information available before that week. Second, the final 104 weeks were reserved as a chronological test set rather than using a random split, preventing the models from training on observations that occur after the forecast period begins. Third, the StandardScaler used for the hourly LSTM was fitted exclusively on the training data and then applied unchanged to the validation and test sets, ensuring that future information did not influence preprocessing. For the feature-based models, the evaluation should be interpreted as a rolling one-week-ahead forecasting exercise because realised load from earlier test weeks became available when constructing lag features for later test weeks, rather than as a strict fixed-origin two-year forecast. In addition, observed contemporaneous temperature was supplied during the test period, making the SARIMAX and feature-based results conditional (explanatory) forecasts rather than fully operational forecasts, since future temperature would not be known exactly at the forecast origin

Q3. For the SARIMAX model, justify the chosen differencing orders and seasonal period.

d = 1 and D = 1 at s = 52 were chosen primarily from the stationarity evidence in Section 2.2 and the STL decomposition in Section 2.1. The first-differenced series passed both ADF and KPSS cleanly, and the STL seasonal strength of 0.871 together with a persistent autocorrelation spike near lag 52 in the level series justified an additional seasonal difference at the annual (52-week) period, which is the natural cycle length for weekly-sampled demand data driven by heating and cooling. The level-series ADF result was ambiguous (it rejected the unit-root null even without differencing), which I judged to reflect limited test power against a strongly seasonal series rather than genuine stationarity, so it was not used as the basis for the differencing decision. I am considerably less confident, by contrast, in the specific autoregressive/moving-average orders (p = 2, q = 6, seasonal (P,Q) = (0,1)) returned by the AIC grid search: as discussed under Q1 and in Section 5.1, this order did not converge cleanly and contains several statistically insignificant coefficients, so while d and D are justified on stationarity grounds, the ARMA orders built on top of them should be read as an AIC artefact rather than a fully validated specification.

Q4. Do temperature and holiday covariates improve forecast accuracy, and are they known at the forecast origin?

Temperature provided a modest improvement in forecast accuracy because weekly electricity demand showed a clear relationship with weather, with correlations of (r=-0.636) for mean temperature and (r=+0.687) for heating degree-days. Incorporating these variables as exogenous regressors in the SARIMAX model reduced MASE from 2.301 to 2.146 and lowered mean forecast bias, indicating that weather information explains part of the variation in electricity demand. However, the improvement was not sufficient to outperform the seasonal naive benchmark, partly because it was built on a SARIMA specification that still exhibited systematic bias. The temperature values used in the test period were the temperatures that were actually observed rather than weather forecasts, meaning the resulting SARIMAX forecasts should be interpreted as conditional or explanatory forecasts rather than fully operational forecasts. Holiday covariates were not included in the final fitted models, so their impact on forecast accuracy cannot be evaluated directly in this study. Nevertheless, unlike temperature, holiday indicators are known with certainty at the forecast origin and therefore represent a potentially valuable exogenous variable that could improve long-horizon forecasting performance in future work.

Q5. Compare the interpretability and complexity of SARIMAX, feature-based, and neural network models.

SARIMAX is the most interpretable in principle, with coefficients that carry a direct statistical meaning and confidence intervals that fall out of the estimation naturally, but Section 4.2 and 5.3 show that this interpretability is only as trustworthy as the underlying estimation: a coefficient table dominated by large standard errors and a convergence warning is not, in practice, very informative, whatever its statistical pedigree. The feature-based models (Random Forest, Gradient Boosting) sit in the middle — feature importances (Figure 8) gave a genuinely useful account of what the model was doing, dominated by the lag-52 feature, but they lack a native uncertainty interval and would need an additional step such as quantile regression forests to produce one. The LSTM was both the least interpretable of the three — its predictions cannot be traced to any individual, meaningful parameter — and, in this study, the least accurate of the nine models tested overall, though its weekly-aggregated error was much closer to the weaker benchmarks (Naive, Mean, Drift) than to Random Forest or Gradient Boosting, which is a useful empirical counterpoint to the assumption that architectural sophistication yields better forecasts by default. In terms of complexity, SARIMAX required the most manual specification work (order selection, exogenous variable choice), the feature-based models required the most feature engineering, and the LSTM required the most computational and hyperparameter-tuning effort, without a matching accuracy payoff here.

Q6. Recommend one model for operational use, and justify the choice using accuracy, uncertainty, interpretability, and maintenance.

I recommend Random Forest. On accuracy, it is the strongest model tested (MASE 1.405) and Gradient Boosting (MASE 1.411) are clearly and consistently beat seasonal naive. On uncertainty, it is currently a weakness — unlike SARIMAX, it does not produce a forecast interval natively — but this is a solvable, well-documented gap (quantile regression forests or residual bootstrapping, Section 7), rather than a fundamental limitation of the model family. On interpretability, its feature-importance output is not as rigorous as a regression coefficient but is considerably more informative in practice than the poorly identified SARIMA/SARIMAX coefficients obtained in this study, and easier to communicate to non-technical stakeholders than an LSTM's internal weights. On maintenance, it is inexpensive and fast to retrain as new weekly data arrives, in sharp contrast to the multi-hour SARIMA grid search or the architecture search, scaling and iterative-rollout machinery the LSTM pipeline requires. SARIMAX remains a useful interpretable cross-check precisely because its assumptions are explicit and auditable, and the LSTM is worth retaining in the toolkit for shorter, higher-resolution forecasting tasks where its capacity for learning intraday structure is more likely to pay off — but for this specific two-year weekly operational forecasting task, Random Forest offers the best balance of the four criteria.

7. Limitations and Future Work

- Model-order selection and architecture: repeat the SARIMA/SARIMAX search with BIC alongside AIC, and replace the LSTM's iterative rollout with a direct multi-step or encoder-decoder sequence-to-sequence architecture (Kong et al., 2019), to address the two main weaknesses identified in Section 6.
- Covariates: enable the prepared-but-unused holiday indicators, and substitute genuine short-range weather forecasts (or seasonal-normal climatology at longer horizons) for the observed Berlin temperature, converting SARIMAX and the feature-based models into genuinely operational forecasts.
- Uncertainty quantification: add quantile regression forests or residual bootstrapping to Random Forest/Gradient Boosting so they produce a forecast interval comparable to SARIMAX's, not only a point forecast.
- Cross-validation: use rolling-origin (walk-forward) evaluation across several two-year windows rather than a single split, to check whether the model ranking in Table 2 is stable or specific to this test period's unusual demand shock.
- Alternative model families: benchmark against Prophet (Taylor and Letham, 2018) and global/cross-series deep-learning forecasters, which have shown strong results on related energy-demand tasks.

8. Conclusion

Across nine forecasting approaches applied to a two-year weekly forecast horizon for German electricity demand, only Random Forest and Gradient Boosting produced a clear, meaningful improvement over the seasonal-naive benchmark, largely by combining an explicit seasonal anchor with smaller local corrections from recent load and temperature. SARIMA and SARIMAX, despite their added structure, underperformed seasonal naive, primarily because AIC-based order selection produced an over-parameterised, poorly identified model; and the LSTM, despite well-behaved training, produced the least accurate forecast of all — though only narrowly behind the Drift benchmark — due to error accumulation in its iterative multi-step rollout rather than any inherent unsuitability of recurrent networks for this task. Based on accuracy, robustness and maintenance cost, Random Forest is the model I would recommend for operational deployment, with the caveat that a proper uncertainty-quantification step, genuine (rather than observed) weather inputs, and holiday covariates should all be added before it is used to inform real operational decisions.

References

Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and Ljung, G.M. (2015) Time Series Analysis: Forecasting and Control. 5th edn. Hoboken, NJ: Wiley.

Breiman, L. (2001) 'Random Forests', Machine Learning, 45(1), pp. 5–32.

Cleveland, R.B., Cleveland, W.S., McRae, J.E. and Terpenning, I. (1990) 'STL: A Seasonal-Trend Decomposition Procedure Based on Loess', Journal of Official Statistics, 6(1), pp. 3–73.

Friedman, J.H. (2001) 'Greedy Function Approximation: A Gradient Boosting Machine', Annals of Statistics, 29(5), pp. 1189–1232.

Hochreiter, S. and Schmidhuber, J. (1997) 'Long Short-Term Memory', Neural Computation, 9(8), pp. 1735–1780.

Hyndman, R.J. and Athanasopoulos, G. (2021) Forecasting: Principles and Practice. 3rd edn. Melbourne: OTexts.

Kong, W., Dong, Z.Y., Jia, Y., Hill, D.J., Xu, Y. and Zhang, Y. (2019) 'Short-Term Residential Load Forecasting Based on LSTM Recurrent Neural Network', IEEE Transactions on Smart Grid, 10(1), pp. 841–851.

Open-Meteo (2024) Historical Weather API. Available at: https://open-meteo.com/en/docs/historical-weather-api (Accessed: 10 July 2026).

Open Power System Data (2020) Time Series Data Package, version 2020-10-06. Available at: https://data.open-power-system-data.org/time_series/ (Accessed: 10 July 2026).